

PocketLedger: Development Specifications and Implementation Prompts

This document provides a comprehensive set of development prompts and technical specifications for **PocketLedger**, a full-stack personal finance and ledger platform. Designed to facilitate the creation of a responsive web application and a desktop-based tracker, these prompts ensure the delivery of a high-quality user experience centered on the core promise: "Track your daily income and expenses."

1. Authentication and Account Management

The authentication module serves as the gateway to the PocketLedger ecosystem, prioritizing security and user convenience. Developers are tasked with implementing a robust account management system that supports traditional credentials alongside modern biometric-equivalent methods.

Feature	Requirement Description
Registration	A form capturing full name, email, and password, featuring real-time password strength evaluation and server-side email verification tracking.
Login	Secure entry via email and password with a "Remember Me" function to enhance returning user experience.
Recovery	A secure forgot-password workflow that triggers a time-limited, token-based reset email from the backend.
Security	Integration of WebAuthn for platform passkeys and an optional 4-digit PIN for rapid re-authentication within active sessions.
Session	JWT-based management with silent refresh capabilities and automatic logout upon refresh-token expiration.

Termination	A formal account deletion process including a confirmation step and permanent data purging.
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2. Financial Dashboard and Spend Analysis

The dashboard is the central hub for financial oversight, transforming raw transaction data into actionable insights. It must provide a high-level summary of the user's financial health through intuitive visualizations and period-based analysis.

Component	Functional Specification
Period Control	A selector for filtering data by Today, This Week, This Month, or custom date ranges.
Balance Summary	A prominent display of the current total balance and savings rate indicators.
Cash Flow	Summary cards for total income and expenses, distinguished by directional icons for inflow and outflow.
Visual Analysis	An interactive donut or pie chart detailing category-wise expense distribution.
Ranking	A list of top spending categories, ranked by amount and accompanied by identifying icons.
Recent Activity	A preview of the latest transactions with navigation to the full history view.

3. Transaction Entry and Management

Efficient transaction logging is critical to user retention. The implementation should focus on a low-friction "Add Transaction" flow that captures necessary metadata while maintaining speed.

Field / Action	Implementation Detail
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Entry Interface	A modal or sheet interface featuring a type toggle (Income vs. Expense) and precise amount input.
Metadata	Selectors for category, payment method (Cash, Card, UPI, Bank Transfer), and wallet (Personal, Business, etc.).
Documentation	A date picker for transaction timing and a free-text note field for additional context.
Receipts	Support for image attachments via camera capture or file upload, including preview and removal options.
Lifecycle	Comprehensive capabilities to edit existing entries or delete them following a confirmation prompt.

4. Categorization and Organization

Categorization allows users to segment their spending patterns. The system must support both pre-defined defaults and user-generated custom categories to ensure maximum flexibility.

Management Task	Technical Requirement
Creation	Tools to define new categories with unique names, icons, and color-coded identifiers.
Modification	Ability to update category attributes as user needs evolve.
Archiving	A mechanism to hide unused categories while preserving their historical transaction data.
Defaults	A curated set of common income and expense categories provided upon initial account setup.

5. Historical Data and Search

The History module provides a granular view of all financial activities. It must support

advanced filtering and search capabilities to allow users to audit their records efficiently.

Feature	Capability Specification
Search	Full-text search across notes, categories, and specific transaction amounts.
Filtering	Multi-select filters for date ranges, transaction types, categories, wallets, and payment methods.
Sorting	Dynamic sorting by date (chronological or reverse) and amount (ascending or descending).
Display	A clean, chronological list view utilizing infinite scrolling or pagination for performance.

6. Reporting and Data Insights

Advanced reporting transforms historical data into trends. Developers should implement a reporting engine capable of generating structured summaries and visual charts.

Report Feature	Requirement
Periodicity	Automated generation of weekly, monthly, and yearly financial summaries.
Visuals	Bar charts for trend analysis and pie charts for distribution overviews.
Export	One-click generation and download of reports in both CSV and PDF formats.
Customization	Parameters to define reports based on specific categories or custom date intervals.

7. System Settings and Customization

The settings module empowers users to tailor the application to their specific needs, from localization to security configurations.

Setting Category	Options and Controls
Localization	Support for multiple currencies, including INR, USD, EUR, and GBP.
Interface	Theme toggles for light and dark modes to suit user preference and environment.
Preferences	Configuration of daily reminders, summary notifications, and budget-overrun alerts.
Data Tools	Access to data backup and restoration utilities, alongside wallet and budget management interfaces.

This specification serves as the primary reference for the development of the PocketLedger application, ensuring all requirements from the PRD are met with technical precision.